# Machine Learning in Digital Forensics: A Systematic Literature Review

Concise Study Summary • Based on the original 2023 paper by Nayerifard et al.

**Important note:** This is a student-friendly summary of the original paper, not a shortened or replacement copy of the authors' publication. The original paper is 99 pages and is available from arXiv.

## 1. Paper details

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## 2. Abstract in simple words

Digital crimes are becoming more complex, and the amount of digital data that investigators must examine is increasing. This makes it difficult to find useful evidence manually. The paper reviews research published from January 2010 to December 2021 on the use of machine learning in digital forensics. It identifies the forensic areas that have received the most attention, the machine-learning methods commonly used, and research gaps that still need work.

## 3. Main objectives

- Review published studies on machine learning applications in digital forensics.
- Identify the main digital-forensics domains where machine learning is used.
- Identify the machine-learning techniques most frequently discussed in the reviewed studies.
- Highlight research gaps and areas where further investigation is needed.

## 4. Key findings

**Image forensics received the greatest attention:** Among the forensic domains reviewed, image forensics benefited most from machine-learning research.

**CNN models were prominent:** Convolutional Neural Network (CNN)-based approaches were identified as especially important and increasingly used in digital forensics.

**Research landscape:** The authors also use keyword-based visual analysis and a comprehensive mind map to present relationships among research themes.

## 5. Machine learning in digital forensics

Machine learning enables a computer system to learn patterns from data and use those patterns to classify, detect, or analyze information. In digital forensics, it can help investigators process large collections of digital material and identify items that may deserve closer examination. It supports the investigator; it does not independently establish that an item is legally valid evidence.

### Examples of relevant forensic tasks

- Image analysis: detecting manipulation or classifying image characteristics.
- Pattern recognition: finding repeated or unusual patterns in digital data.
- Evidence triage: helping prioritize large amounts of material for human review.

- Research mapping: organizing methods and application areas reported in earlier studies.

## 6. Important terms

**Digital Forensics:** The process of identifying, collecting, preserving, examining, and presenting digital evidence.

**Machine Learning (ML):** A branch of AI in which systems learn patterns from data.

**CNN:** A neural-network approach commonly used to analyze image data.

**Systematic Literature Review:** A structured method for finding, selecting, and analyzing previously published studies.

**Research gap:** A question or area that existing research has not adequately addressed.

## 7. Challenges and research gaps

The review emphasizes that the field still needs further research. Practical challenges for machine-learning-based forensic work include the availability and quality of datasets, the reliability and generalization of models, explainability of model decisions, and the need for careful validation. These are important considerations when applying ML to real investigations; model output should be checked by qualified human investigators.

## 8. Conclusion

The paper shows that machine learning has growing importance in digital forensics. Image forensics is the most actively explored area in the reviewed literature, and CNN-based methods are particularly prominent. The review helps readers understand the research landscape and points toward areas that need more study. Machine learning can make evidence analysis more scalable, but accuracy, validation, transparency, and expert oversight remain essential.

## 9. Short explanation for viva / teacher

"This research paper reviews how Machine Learning is used in Digital Forensics. The authors studied research papers published between 2010 and 2021. They found that image forensics received the greatest benefit from machine-learning methods and that CNN-based models are widely used. The paper also identifies research gaps and explains why further work is needed to make digital investigations more effective and reliable."

## 10. Reference and original source

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